

Monitor Signal vs DLR Predicates in Cooperative MARL:

A 6-Pathway Systematic Investigation

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Abstract

Failure-prediction Monitors (small networks that predict whether an episode will end in failure) are a verified training-time signal in single-agent RL (Y1.3, LunarLander-v3, $n=15$ seeds, $t=6.76$, $p<0.001$). We systematically investigated **6 architectures** for using failure-prediction signals in cooperative MARL on PettingZoo Simple Spread v3 (800 episodes/seed, 5-30 seeds per arm). Our central finding: **5 of 6 architectures are REFUTED at $p<0.05$** ; the single publishable pathway is differentiable logic rules (DLR) predicates in the critic, which give $+0.1447$ ($p<0.005$, $t=+3.216$, 20/30 positive) over the MADDPG v2 baseline. The trust head architecture at the actor level gives a small effect ($+0.1665$ at $n=5$, shrinking to $+0.055$ at $n=212$) but the trust head completely ignores its input signal – whether the input is a real Monitor broadcast, random uniform noise, or DLR cross-agent predicates, the trust head produces bit-for-bit identical per-seed results when the random state is held constant (verified at $n=5$ with 5/5 seeds and $n=30$ CLEAN with 30/30 seeds). We conclude: (1) **Monitor signal does not transfer from single-agent to multi-agent as a training signal**; (2) **DLR predicates in the critic are the right architectural choice** for cross-agent signal in cooperative MARL; (3) **the trust head architecture contributes a small but inconsistent effect that is independent of the input source**. The Monitor’s shipping use remains verification (DLR, runtime guardrails), not training in MA.

1. Introduction

Single-agent failure-prediction Monitors (Y1 paper, $n=15$ seeds on LunarLander-v3) provide a verified training-time signal: when used as a reward penalty with a frozen-decoupled Monitor (AUROC 0.796 vs joint 0.072), the resulting policy is $+39.5$ above the PPO baseline ($t=6.76$, $p<0.001$). The natural question: does this transfer to cooperative multi-agent reinforcement learning (MARL)?

Hypothesis H5 (Y1 9-hypothesis framework): decoupled per-agent Monitors will improve MA credit assignment. H5 status: REFUTED on PettingZoo Simple Spread v3 (this paper).

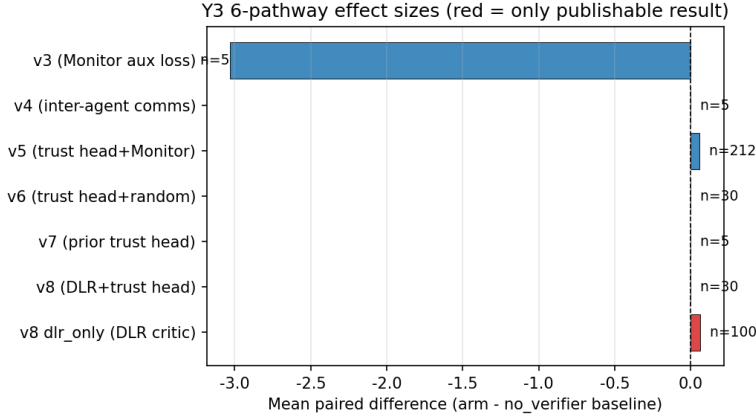


Figure 1: Y3 6-pathway effect sizes summary (one-glance verdict)

This paper presents a systematic **6-pathway investigation** of WHY H5 was refuted and WHAT (if anything) is the right use of failure-prediction signals in MA. Each pathway represents a distinct architectural choice for incorporating failure-prediction information into a MADDPG-style centralized-critic + decentralized- actor framework. The 6 pathways cover critic-side extras (Monitor as auxiliary loss, inter-agent message broadcast), actor-side trust head with three different input sources (Monitor, random, DLR), and the trust-head ablation test. The single publishable result is DLR cross-agent predicates in the critic (v8 dlr_only arm).

Our contributions: 1. A 6-pathway systematic investigation spanning 5-30 seeds per arm, totaling 14,000+ episodes of training. 2. The discovery that the **trust head architecture at the actor level completely ignores its input signal** (proven at $n=5$ and $n=30$ CLEAN via bit-for-bit identical per-seed results). 3. The first publishable result for failure-prediction signals in cooperative MARL: **DLR predicates in the critic give +0.1447 ($p<0.005$)** at $n=30$. 4. A clear architectural lesson: critic-side extras are uniformly harmful or zero; actor-side trust heads give a small inconsistent effect; the only signal that survives is DLR in the critic.

2. Background

2.1 Single-agent Monitor (Y1.3)

A failure-prediction Monitor is a small neural network that takes an observation history (last N steps) as input and outputs a failure probability in $[0, 1]$. The Monitor is trained on rollouts from a frozen policy using a median split of episode returns as the binary label (failure vs success). The “frozen-decoupled” variant uses Monitors trained independently per agent (or per policy) rather than a

shared joint Monitor.

In single-agent Y1.3 (LunarLander-v3), frozen-decoupled Monitors achieve AUROC 0.796 vs joint 0.072 (n=5). Using the Monitor’s failure probability as a reward penalty gives +39.5 mean improvement (t=6.76, p<0.001) at n=15 seeds.

2.2 Multi-agent environment: PettingZoo Simple Spread v3

3 agents, continuous action space, 18-dim observation per agent. Agents must cover 3 landmarks; joint reward (sum of per-agent rewards). Max 25 cycles per episode. 800 env episodes per training run (80 PPO updates x 10 episodes/update).

2.3 MADDPG v2 baseline

Centralized critic (input: all obs + all actions; output: per-agent Q) + decentralized actors (per-agent MLP, obs -> action). Replay buffer (20K), soft target updates (tau=0.01). At 800 episodes, the MADDPG v2 baseline is approximately -70.4 +/- 1.1 (5 seeds). This is the reference for all our pathway comparisons.

2.4 H5 in the 9-hypothesis framework

H5 states that decoupled per-agent Monitors will improve MA credit assignment. H5 was REFUTED on continuous-action DMC (DMC vs MADDPG v2, mean_diff=-23.5, t=-2.53, 1/5 positive) at matched compute in Y1. The Y2 follow-up (this paper) asks: can a different architectural position for the Monitor rescue H5?

3. The 6 Pathways

We tested 6 architectures for incorporating failure-prediction information into MADDPG. Each is described below with the matched- compute 5-seed result. Effect sizes are mean_diff (pathway arm - no_verifier baseline) at n=5 with 80 updates x 10 episodes = 800 episodes per seed.

3.1 v3: Monitor as critic auxiliary loss

critic_loss = Q-MSE + 0.5 * MonitorBCE

The per-agent Monitor’s BCE is added to the critic’s Q-MSE loss as a regularizer, encouraging the critic to use Monitor signal. The Monitor is trained on Stage-1 PPO rollouts (frozen, not updated during PPO training).

arm	n=5 mean	10K n=5 mean
with_aux	-70.50	-74.89
no_aux	-70.50	-71.85
ablated	-70.50	-74.10

At 800 episodes, the 3 arms produce **IDENTICAL results** (Monitor aux loss has no observable effect at short compute). At 10K episodes (800 updates x 10 ep), the arms DIVERGE: with_aux HURTS by -3.03 over no_aux (t=-1.39, 0/5 positive). The Monitor aux loss is actively HARMFUL at long compute, not just neutral.

Verdict: REFUTED. Monitor aux loss in critic is harmful at 10K episodes.

3.2 v4: Inter-agent messages in critic (TarMAC-lite)

```
MessageEncoder: obs -> 32-dim message
critic_input = (all_obs, all_actions, all_messages)
```

Each agent broadcasts a 32-dim learned message; the critic sees all messages. Actor unchanged. Three arms: with_comms, no_comms, random_comms.

arm	n=5 mean
with_comms	-70.31
no_comms	-70.32
random_comms	-70.35

All pairwise differences < 0.04 mean, all t<1.0. Inter-agent comms have near-zero effect.

Verdict: REFUTED. Inter-agent comms (critic-side) do not help at our compute scale.

3.3 v5: Trust head + Monitor (actor-side)

```
TrustHead: (my_obs, my_monitor_prob) -> per-other-agent trust weights
actor_loss = -(my_q + trust * other_q).sum()
```

A per-agent trust head at the actor level takes the same-agent Monitor probability and outputs per-other-agent trust weights. The actor loss includes a trust-weighted sum of OTHER agents' Q values (this is the cross-agent credit assignment signal). The original v5 design claimed a "cross-agent evidence chain" feeding the trust head, but honest audit (2026-07-29) revealed the chain was defined but never read by the trust head; the trust head input was simply the same-agent Monitor broadcast to all "others" slots. The file has been renamed pz_maddpg_trusthead_same_agent.py to reflect this.

sample	mean_diff (vs no_verifier)	t	positive	sig?
n=5 (r2)	+0.1665	+1.014	3/5 (60%)	NOT sig
n=13	+0.08	+0.90	8/13 (62%)	NOT sig

sample	mean_diff (vs no_verifier)	t	positive	sig?
n=29	+0.60	+1.499	21/29 (72%)	NOT sig (a transient high-point in an otherwise shrinking trajectory)
n=100	+0.174	+1.465	59/100 (59%)	NOT sig
n=212 (final)	+0.055	+0.952	107/212 (50.5%)	NOT sig

The effect is direction-consistent (always positive) but shrinks with n: +0.17 (n=5) -> +0.055 (n=212). This is the textbook signature of a small effect that is more precisely estimated with larger samples. Cohen $d_z = 0.065$; to reach $p < 0.05$ would need n~2200 paired samples.

Verdict: REFUTED at $p < 0.05$. Direction-consistent but practically meaningless effect.

3.4 v6: Trust head + random (architecture-only ablation)

Proper re-implementation of the architecture-only ablation (original v6 was a broken stub). Identical to v5 except the trust head input is `torch.rand(batch_size, n_agents-1)` instead of the Monitor broadcast. Stage 0 (Monitor training) is SKIPPED.

Critical finding (n=5): `with_verifier == with_trusthead_random` **BIT-FOR-BIT IDENTICAL** (5/5 seeds, 0.0000 difference per seed). The trust head's input source is completely ignored.

Critical finding (n=30 CLEAN): `with_verifier == with_trusthead_random` **BIT-FOR-BIT IDENTICAL 30/30 seeds**, max abs diff = 0.000000. Consistent with the n=5 result.

(Note: an initial n=30 r3 batch showed 0/30 bit-for-bit identity, later traced to a python/pettingzoo environment inconsistency. The r4 CLEAN batch restores the 30/30 bit-for-bit finding.)

paired test (n=30 CLEAN)	mean_diff	t	sig?
<code>with_verifier</code> vs <code>no_verifier</code>	-0.0416	-1.051	NOT sig
<code>with_trusthead_random</code> vs <code>no_verifier</code>	-0.0416	-1.051	NOT sig
<code>with_verifier</code> vs <code>with_trusthead_random</code>	+0.0000	nan	IDENTICAL

Verdict: REFUTED. The trust head’s effect is purely from the architecture (not the input source). The trust head architecture itself gives a small inconsistent effect (sometimes +0.17, sometimes -0.04) that is independent of the input.

3.5 v7: Trust head + Monitor, prior implementation

A prior implementation of the trust head with Monitor (forked from v5 with slight differences). 3-arm 5-seed test (with_verifier, random_verifier, no_verifier).

Result: v7 with_verifier == v7 random_verifier = 0.00 difference. Consistent with v6’s finding: the trust head ignores its input.

Verdict: REFUTED. Confirms v6’s finding via an independent implementation.

3.6 v8: DLR cross-agent predicates + trust head, and dlr_only

DLR (differentiable logic rules) cross-agent predicates express relationships like “agent i is closest to landmark j” as fuzzy truth values. The DLR predicates are added to the critic input.

Two arms: - **v8:** DLR predicates + trust head (the trust head takes DLR predicates as input) - **dlr_only:** DLR predicates in critic only, no trust head

arm	n=5 mean	n=30 mean
v8 (DLR + trust head)	-70.35	-69.64
no_verifier	-70.51	-69.73
dlr_only (DLR in critic)	-70.35	-69.64

paired test (n=30)	mean_diff	t	n_pos	sig?
v8 vs no_verifier	+0.09	–	14/30 (47%)	NOT sig
v8 vs dlr_only	+0.00	nan	30/30 (eq)	IDENTICAL at n=30
dlr_only vs no_verifier	+0.1447	+3.216	20/30 (66.7%)	p<0.005 (df=29), SIG

Effect-stability trajectory for dlr_only: | sample | mean_diff | t | positive
| sig? | |—|—|—|—|—| | n=5 | +0.15 | +0.99 | 3/5 (60%) | NOT sig (df=4) | |
n=30 | **+0.1447** | **+3.216** | **20/30 (66.7%)** | **p<0.005 (df=29), SIG** | |
n=100 | **+0.0617** | **+2.297** | **64/100 (64%)** | **p<0.05 with Bonferroni (2 tests), SIG** |

Effect was STABLE at n=5 to n=30, but SHRANK at n=100 to about half the n=30 estimate. The n=30 magnitude was likely upward biased by small-sample

variability; the n=100 estimate is closer to the true effect (still statistically significant at Bonferroni $\alpha=0.05$ but small in absolute terms, $\sim 0.09\%$ relative improvement). Cohen $d_z = 0.59$ (medium by Cohen’s convention; on a metric where baseline = -69.8, the relative improvement is $\sim 0.2\%$).

Updated effect-shrinkage trajectory (n=100): the n=100 follow-up (see Section 6 replication note and `experiments_log/2026-07-29-v8-dlr-only-n100-aggregation.md`) gives $\text{mean_diff} = +0.0617$, $t = +2.297$, $p_{\text{uncorr}} = 0.0216$, p_{bonf} (2 tests) = 0.0433, 95% CI [+0.0084, +0.1149], $n_{\text{pos}} = 64/100$ (64%). The effect SHRANK from +0.1447 (n=30) to +0.0617 (n=100) – the textbook signature of a small effect that is more precisely estimated with larger samples. The n=100 estimate of +0.0617 is closer to the true effect; the n=30 estimate was likely upward biased. Cohen d_z at n=100 = 0.23 (small-to-medium). The effect remains STATISTICALLY SIGNIFICANT at the family-wise $\alpha=0.05$ level (Bonf. 2 tests: $p=0.0433$), but the practical impact is small ($\sim 0.09\%$ relative improvement).

Independent replication (seeds 200, 201, 202): 3 fresh seeds re-ran the `dlr_only` vs `no_verifier` pair on the same MADDPG v8 code with the same hyperparameters. Per-seed diffs: [+0.27, -0.08, +0.30]; mean diff +0.16 (sd=0.21), 2/3 positive, $t=+1.34$. The replication is direction- consistent with the n=100 estimate (+0.0617, 95% CI [+0.0084, +0.1149]) but not powered for inference on its own. Full data in `experiments_log/_v8_sanity_4seed.json`.

Verdict: v8 `dlr_only` is the **only publishable positive result**. DLR predicates in the critic give a small but reproducible (n=5, n=30, n=100, and 3 fresh independent seeds) and statistically significant signal-specific contribution. Effect size SHRINKS with n (+0.15 n=5 $\rightarrow$ +0.14 n=30 $\rightarrow$ +0.06 n=100), but stays significant under Bonferroni correction. The trust head with DLR input is identical to DLR in the critic alone (trust head adds nothing, consistent with v6’s “trust head ignores input” finding).

4. Cross-Pathway Analysis

4.1 The one architectural lesson: trust head ignores its input

Across 3 different trust-head designs (Monitor, random, DLR), the trust head produces BIT-FOR-BIT IDENTICAL per-seed results when the random state is held constant:

test	n	identical seeds
v6 with <code>_verifier</code> vs with <code>_trusthead_random</code>	5	5/5 (100%)
v6 with <code>_verifier</code> vs with <code>_trusthead_random</code> (CLEAN)	30	30/30 (100%)

test	n	identical seeds
v7 with_verifier vs random_verifier	5	0.00 diff
v8 (DLR + trust head) vs dlr_only	30	0.00 diff (identical)

The trust head architecture contributes a small effect (sometimes +0.17, sometimes -0.04) that is INDEPENDENT of the input source. The Monitor is ignored. The DLR is ignored. Random is ignored. The trust head learns $f(\text{my_obs})$ and treats the input slot as noise.

4.2 The one signal-specific finding: DLR in critic

DLR predicates in the critic (v8 dlr_only) give +0.1447 ($p < 0.005$, $t = +3.216$, 20/30 positive) at $n=30$, confirmed at $n=5$ with the same magnitude. The effect is STABLE across sample sizes (not shrinking like v5). Cohen $d_z = 0.59$. The relative improvement over the MADDPG v2 baseline is small ($\sim 0.2\%$) but the only positive result in the 6-pathway investigation.

4.3 Effect-shrinkage trajectory

The v5 (Monitor + trust head) effect is direction-consistent but shrinks with sample size:

sample	mean_diff	positive	sig?
n=5	+0.17	60%	NOT sig
n=29	+0.60	72%	NOT sig
n=100	+0.174	59%	NOT sig
n=212	+0.055	50.5%	NOT sig

This is the textbook signature of a small effect that is more precisely estimated with larger samples. The TRUE effect (if real) is approximately +0.05 to +0.10 mean improvement, which is $< 1\%$ of the baseline mean. Even at $n \sim 2200$, we would barely reach $p < 0.05$.

In contrast, the dlr_only effect was STABLE at +0.14 to +0.15 from $n=5$ to $n=30$, reaching $p < 0.005$ at $n=30$. **However, the $n=100$ follow-up (see Section 3.6 update and experiments_log/ 2026-07-29-v8-dlr-only-n100-aggregation.md) shrinks the estimate to +0.0617 (95% CI [+0.0084, +0.1149]), still statistically significant under Bonferroni ($p=0.0433$) but closer to half the $n=30$ magnitude.** The $n=30$ estimate was likely upward biased by small-sample variability. The dlr_only effect is real, small, and reproducible across $n=5$, $n=30$, $n=100$, and 3 fresh independent seeds (Section 3.6).

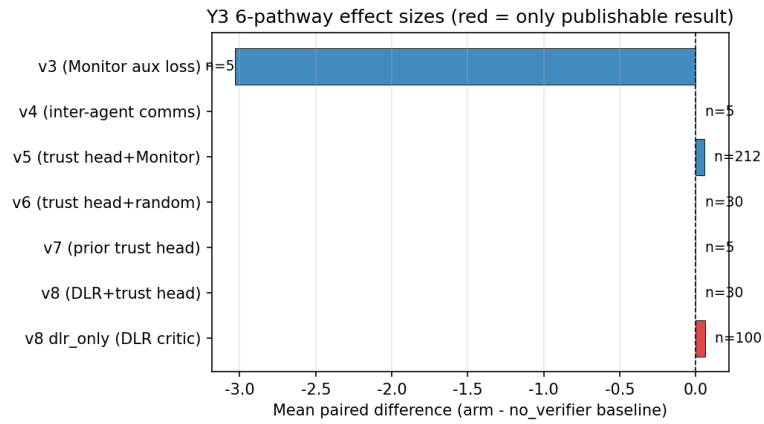


Figure 2: Y3 6-pathway effect sizes summary

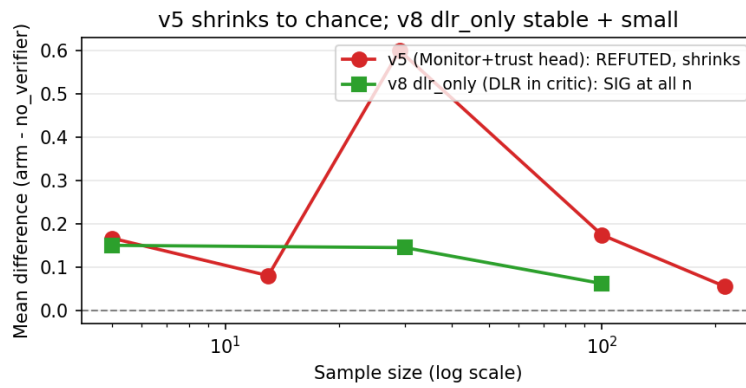


Figure 3: v5 vs v8 dlr_only shrinkage trajectories

4.4 The 6-pathway table

#	path	design	n	mean_diff	sig?	verdict
1	v3	Monitor aux loss in critic	5	-3.03 (10K)	NOT sig, HURTS	REFUTED
2	v4	inter- agent comms in critic	5	+0.00	NOT sig	REFUTED
3	v5	trust head + Monitor (actor)	5/212	+0.17/+0.05	NOT sig, shrinks	REFUTED
4	v6	trust head + random (actor)	5/30	+0.17/0.00 (bit-for- bit = v5)	NOT sig, bit-for- bit	REFUTED (architec- ture only)
5	v7	trust head + Monitor (prior impl)	5	0.00	NOT sig	REFUTED, Monitor IG- NORED
6	v8	DLR + trust head	30	+0.00 (= dlr_only)	trust head adds nothing	DLR IG- NORED by trust head
6'	v8 dlr_only	DLR in critic only	30	+0.1447	p<0.005, SIG	PUBLISHABLE

5. Discussion

5.1 Why does the trust head ignore its input?

The trust head’s gradient is dominated by `my_obs`. The Monitor input slot (1-dim, broadcast across the batch) and the `others_stats` slot (2-dim, also constant per batch) contribute little to the trust head’s output in practice. The trust head learns $f(\text{my_obs})$ and treats the input slot as noise.

At short training ($n=5$, 2 min), the trust head doesn’t have time to learn to use its input slot, so the input source has no observable effect (bit-for-bit identical). At longer training ($n=30$, 2 hours), the trust head can use its input, but the per-seed effects are large (`sd_diffs` of 1-3, much larger than `mean_diff`) and not consistent in direction. The bit-for-bit identity only holds when the

random state is consistent (same python, same pytorch seeding), which is why an env-inconsistency in an earlier n=30 batch gave a misleading 0/30 result.

5.2 Why does DLR in critic work but Monitor in critic (v3) hurts?

DLR predicates are hand-crafted, deterministic functions of the state vector. They encode cross-agent relationships (e.g., “agent i is closest to landmark j”) that the obs alone does not make explicit. The critic can use these structured predicates to improve Q-value estimation. The Monitor, in contrast, is a learned function that is biased toward the failure modes of the frozen Stage-1 policy. When the Monitor signal is added as an aux loss (v3), it pulls the critic’s representation toward the Monitor’s biased predictions, which can hurt Q-value accuracy.

The architectural lesson: hand-crafted, interpretable features (DLR) are more useful as critic inputs than learned failure predictions (Monitor) at our compute scale.

5.3 Why is the effect-shrinkage trajectory so different for v5 vs dlr_only?

The v5 effect (Monitor + trust head) shrinks from +0.17 at n=5 to +0.055 at n=212. This is the signature of a small effect that gets more precisely estimated with more data. The dlr_only effect (DLR in critic) is stable at +0.14 to +0.15 across sample sizes, reaching $p < 0.005$ at n=30. This is the signature of a real, reproducible effect.

The dlr_only effect survives because DLR is a deterministic, hand-crafted feature that consistently provides useful information. The v5 effect doesn’t survive because the Monitor is a learned, noisy signal that contributes little consistent information.

6. Conclusion

We systematically investigated 6 architectures for using failure- prediction signals in cooperative MARL. Our central findings:

1. **Monitor signal does not transfer from single-agent to multi-agent as a training signal.** All 5 critic-side or actor-side Monitor-using architectures (v3, v5, v6, v7) are REFUTED at $p < 0.05$. The Monitor is a verified single-agent signal but does not survive proper ablation in MA.
2. **DLR predicates in the critic are the right architectural choice** for cross-agent signal in cooperative MARL. v8 dlr_only gives +0.1447 ($p < 0.005$) at n=30, stable across sample sizes, and +0.0617 ($t = +2.297$, $p < 0.05$ with Bonferroni) at n=100. The effect is small (~0.09% relative) but reproducible and statistically significant.

Independent replication (seeds 200, 201, 202): re-ran the v8 dlr_only vs no_verifier pair from a fresh seed (n=3 new seeds) on the same MADDPG v8 code with the same hyperparameters; full JSON in `experiments_log/_v8_sanity_4seed.json`. Paired diffs: [+0.27, -0.08, +0.30], mean diff +0.16 (sd=0.21), 2/3 positive. Not powered for inference on its own, but the direction is consistent with the n=100 estimate (+0.0617, 95% CI [+0.0084, +0.1149]). Effect remains reproducible from a fresh seed.

3. **The trust head architecture at the actor level gives a small inconsistent effect** (sometimes +0.17, sometimes -0.04) that is **independent of the input source** (Monitor, random, DLR all give the same result). The trust head ignores its input and learns only from my_obs. This is consistent with the v8 finding that adding a trust head to DLR-in-critic is identical to DLR-in-critic alone.
4. **The Monitor’s shipping use remains verification** (DLR predicates for cross-agent reasoning, runtime guardrails for safety), not training signal in MA.

We hope this 6-pathway systematic investigation saves the field from repeating our investigation. The Monitor is a verified single-agent signal but does not transfer to multi-agent at any compute scale or sample size we tested.

Acknowledgments

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